# Hydrogen-Universe Dual 3D MUGE: Scale Invariance from Bohr Radius to Hubble Horizon

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## PAPER_1105: Hydrogen-Universe Dual 3D MUGE: Scale Invariance from Bohr Radius to Hubble Horizon


### Abstract

We construct a dual 3D Modified Universal Gravity Equation (MUGE) system comparing the hydrogen atom (quantum scale, $r = a_0$) to the observable universe (cosmological scale, $r = R_H$), demonstrating UQFF scale invariance through the 9-term compressed MUGE framework evaluated on volumetric grids at both scales.


### 1. The 9-Term Compressed MUGE

$$g_{\text{MUGE}}(r) = g_N + g_{\text{exp}} + g_{\text{super}} + g_{\text{env}} + g_{\text{Ug}} + g_{\text{cosm}} + g_Q + g_{\text{fluid}} + g_{\text{pert}}$$

#### Term Definitions

| # | Term | Expression | Physics |
|---|---|---|---|
| 1 | $g_{mDPM}$ | $\mu_s \nabla(M_s/r)$ | DPM-seeded gravity |
| 2 | $g_{\text{exp}}$ | $H_0^2\, r$ | Hubble expansion |
| 3 | $g_{\text{super}}$ | $B^2/(2\mu_0 \rho_{\text{avg}}\, r)$ | Magnetic suppression |
| 4 | $g_{\text{env}}$ | $\kappa\, g_N \exp(-1/100)$ | Envelope dissipation |
| 5 | $g_{\text{Ug}}$ | $\sum_{i=1}^{26} Q_i\, g_N/26$ | 26-layer Ug aggregate |
| 6 | $g_{\text{cosm}}$ | $(\Lambda c^2/3)\, r$ | Cosmological constant |
| 7 | $g_Q$ | $\hbar/(M r^3)$ | Quantum correction |
| 8 | $g_{\text{fluid}}$ | $\nu/r^2$ | Navier-Stokes viscous |
| 9 | $g_{\text{pert}}$ | $(4\pi/3) G \rho_{\text{DM}}\, r$ | Dark matter perturbation |

### 2. Hydrogen Scale ($r = a_0$)

Parameters: $M = m_p = 1.673 \times 10^{-27}$ kg, $r = a_0 = 5.292 \times 10^{-11}$ m.

#### Dominant Terms

$$g_N^{(H)} = \frac{G m_p}{a_0^2} \approx 3.99 \times 10^{-8}\ \text{m/s}^2$$

$$g_Q^{(H)} = \frac{\hbar}{m_p\, a_0^3} \approx 4.25 \times 10^{24} \text{ m/s}^2$$

The quantum correction dominates by $\sim 32$ orders of magnitude at the Bohr radius, confirming that atomic-scale MUGE is quantum-dominated.

## 3. Universe Scale ($r = R_H$)

Parameters: $M = M_U = 1.5 \times 10^{53}$ kg, $r = R_H = 4.4 \times 10^{26}$ m.

### Dominant Terms

$$g_N^{(U)} = \frac{G M_U}{R_H^2} \approx 5.17 \times 10^{-5} \text{ m/s}^2$$

$$g_{\text{cosm}}^{(U)} = \frac{\Lambda c^2}{3} R_H \approx 1.44 \times 10^{-5} \text{ m/s}^2$$

At the Hubble radius, DPM-seeded and cosmological-constant terms are comparable, with the Hubble expansion term $g_{\text{exp}}^{(U)} = H_0^2 R_H \approx 2.13 \times 10^{-9}$ m/s$^2$ subdominant.

## 4. Scale Invariance Ratio

$$\mathcal{R} = \frac{g_{\text{MUGE}}^{(H)}}{g_{\text{MUGE}}^{(U)}}$$

This ratio encodes the full UQFF scale-invariance structure. The fact that both systems are described by the same 9-term equation with only mass and radius as inputs demonstrates MUGE universality.

## 5. 3D Volumetric Grid

A $5^3 = 125$-point spatial grid is constructed at each scale, evaluating $g_{\text{MUGE}}$ at each point $\mathbf{r}_{\text{grid}}$ centered on the system. Grid statistics (average, standard deviation) characterize the spatial profile of the gravitational field at each scale.

## 6. Implementation

Calculator: `HydrogenUniverseDual3DMUGECalculator` in CondensedPhysics.py

- Default $5^3$ grid (configurable via `n_grid`)
- Outputs: single-point and grid-averaged MUGE for both scales, scale ratio, per-term breakdown

## 7. Conclusion

The dual hydrogen-universe 3D MUGE system demonstrates that the UQFF 9-term framework spans 37 orders of magnitude in length scale ($10^{-11}$ to $10^{26}$ m), transitioning smoothly from quantum-dominated physics to cosmological-constant-dominated physics through a single unified equation set.

## References


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### Key References with arXiv/DOI Identifiers


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